

11

Nucleophilic substitution at C=O with loss of carbonyl oxygen

Connections

➡ Building on

- Nucleophilic attack on carbonyl groups [ch6](#)
- Acidity and pK_a [ch8](#)
- Nucleophilic substitution at carbonyl groups [ch10](#)

Arriving at

- Replacement of carbonyl oxygen
- Acetal formation
- Imine formation
- Stable and unstable imines
- The Strecker and Wittig reactions

➡ Looking forward to

- Rate and pH [ch12](#)
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- Acylation of enolates [ch26](#)
- Synthesis of alkenes [ch27](#)

Introduction

Nucleophiles add to carbonyl groups to give compounds in which the trigonal carbon atom of the carbonyl group has become tetrahedral.



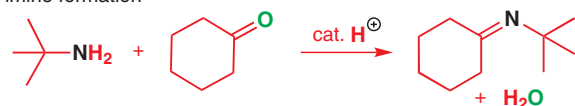
In Chapter 10 you saw that these compounds are not always stable: if the starting material contains a leaving group, the addition product is a **tetrahedral intermediate**, which collapses with loss of the leaving group to give back the carbonyl group, with overall substitution of the leaving group by the nucleophile.



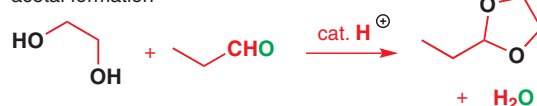
■ Acetals had walk-on parts in Chapters 2 and 6; in this chapter they are one of the stars. They are simply compounds with two oxygen atoms bound to the same saturated carbon atom. This example is cyclic, but others are not, for example $\text{CH}_2(\text{OMe})_2$.

In this chapter you will meet substitution reactions of a different type. Instead of losing a leaving group, the carbonyl group loses its oxygen atom. Here are two important examples: the carbonyl oxygen atom has been replaced by a nitrogen atom during imine formation and by two atoms of oxygen during acetal formation. Notice too the acid catalyst—we shall see shortly why it is required. These are examples of *nucleophilic substitution at the carbonyl group with loss of carbonyl oxygen*.

imine formation



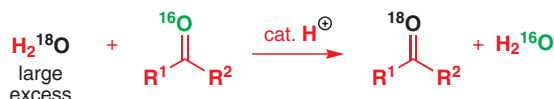
acetal formation



Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

You have, in fact, already met some less important reactions in which the carbonyl oxygen atom can be lost, but you probably didn't notice at the time. The equilibrium between an aldehyde or ketone and its hydrate (p. 134) is one such reaction.

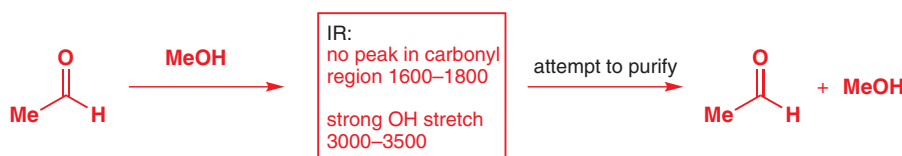
When the hydrate reverts to starting materials, either of its two oxygen atoms must leave: one came from the water and one from the carbonyl group, so 50% of the time the oxygen atom that belonged to the carbonyl group will be lost. Usually, this is of no consequence, but it can be useful. For example, in 1968 some chemists studying the reactions that take place inside mass spectrometers needed to label the carbonyl oxygen atom of a ketone with the isotope ^{18}O .



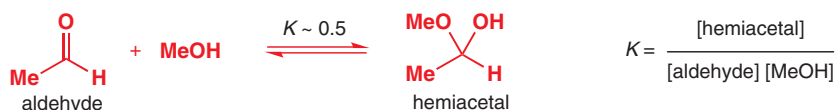
By stirring the 'normal' ^{16}O compound with a large excess of isotopically labelled water for a few hours in the presence of a drop of acid they were able to make the required labelled compound. Without the acid catalyst, the exchange is very slow. Acid catalysis speeds the reaction up by making the carbonyl group more electrophilic so that equilibrium is reached more quickly.

Aldehydes can react with alcohols to form hemiacetals

When acetaldehyde is dissolved in methanol, a reaction takes place: we know this because the IR spectrum of the mixture shows that a new compound has been formed. Most dramatically, the carbonyl frequency is no longer there. However, isolating the product is impossible: it decomposes back to acetaldehyde and methanol.



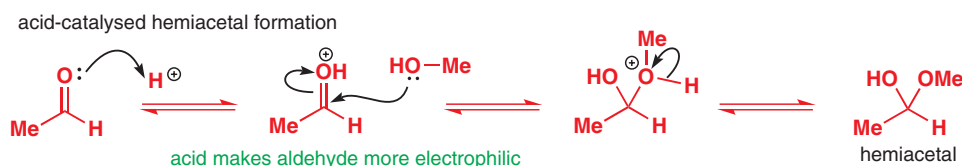
The product is in fact a hemiacetal. Like hydrates, most hemiacetals are unstable with respect to their parent aldehydes and alcohols, for example the equilibrium constant for reaction of acetaldehyde with simple alcohols is about 0.5.



So by making [MeOH] very large (using it as the solvent, for example) we can turn most of the aldehyde into the hemiacetal. However, if we try to purify the hemiacetal by removing the methanol, more hemiacetal keeps decomposing to maintain the equilibrium constant. That is why we can never isolate such hemiacetals in a pure form.

Acid or base catalysts increase the rate of equilibration of hemiacetals with their aldehyde and alcohol parents

Acyclic hemiacetals form relatively slowly from an aldehyde or ketone plus an alcohol, but their rate of formation is greatly increased either by acid or by base. As you would expect from Chapters 6 and 10, acid catalysts work by increasing the electrophilicity of the carbonyl group.

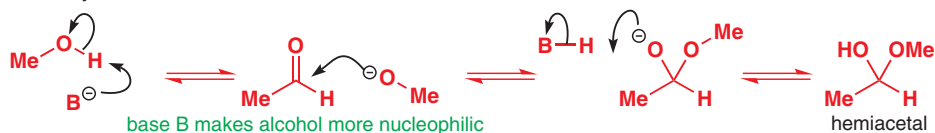

$$\text{H}_2\text{O} + \text{R}^1\text{C}(=\text{O})\text{R}^2 \rightleftharpoons \text{R}^1\text{C}(\text{OH})_2\text{R}^2$$

➔ You met the mechanism for this reversible reaction in Chapter 6.

■ The exceptions are cyclic hemiacetals, as you saw in Chapter 6, in which the nucleophilic OH group is in the same molecule as the electrophilic carbonyl. We will explain how *entropy* accounts for this in Chapter 12.

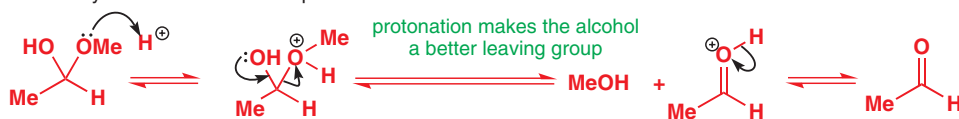
Base catalysts, on the other hand, work by increasing the nucleophilicity of the alcohol by removing the OH proton before it attacks the C=O group. In both cases the energy of the starting materials is raised: in the acid-catalysed reaction the aldehyde is destabilized by protonation and in the base-catalysed reaction the alcohol is destabilized by deprotonation.

base-catalysed hemiacetal formation

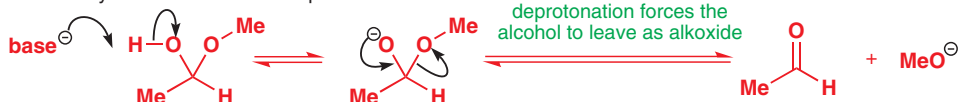


You can see why hemiacetals are unstable: they are essentially tetrahedral intermediates containing a leaving group and, just as acid or base catalyses the formation of hemiacetals, acid or base also catalyses their decomposition back to starting aldehyde or ketone and alcohol. That's why the title of this section indicated that acid or base catalysts increase the rate of equilibration of hemiacetals with their aldehyde and alcohol components—catalysts never change the position of that equilibrium!

acid-catalysed hemiacetal decomposition

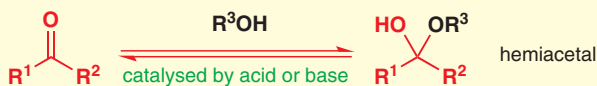


base-catalysed hemiacetal decomposition



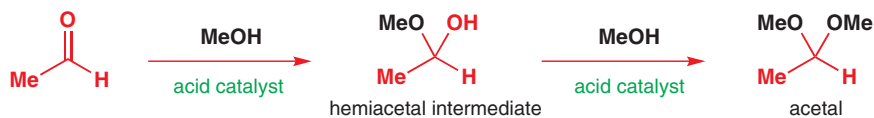
● To summarize

Hemiacetal formation and decomposition are catalysed by acid or base.



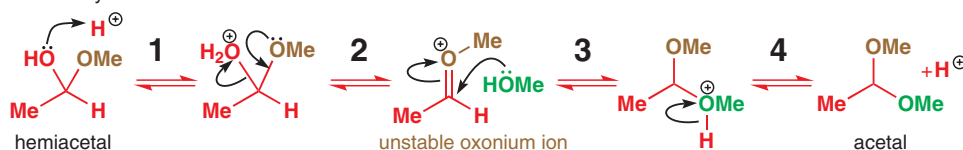
Acetals are formed from aldehydes or ketones plus alcohols in the presence of acid

We said that a solution of acetaldehyde in methanol contains a new compound: a hemiacetal. We've also said that the rate of formation of hemiacetals is increased by adding an acid (or a base) catalyst to the alcohol plus aldehyde mixture. But, if we add catalytic acid to our acetaldehyde–methanol mixture, we find not only that the rate of reaction of the acetaldehyde with the methanol increases, but also that a different product is formed. This product is an acetal; the hemiacetal is half-way there.



In the presence of acid (but not base!) hemiacetals can undergo an elimination reaction (different from the one that just gives back aldehyde plus alcohol), losing the oxygen atom that once belonged to the parent aldehyde's carbonyl group.

acid-catalysed acetal formation from hemiacetal

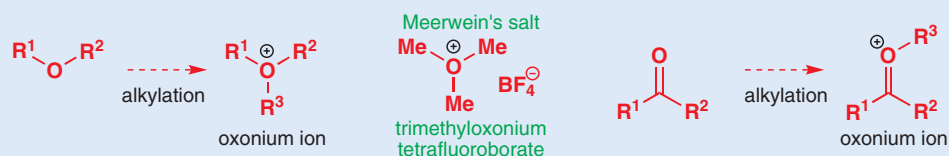


The stages are:

1. Protonation of the hydroxyl group of the hemiacetal.
2. Loss of water by elimination. This elimination leads to an unstable and highly reactive oxonium ion.
3. Addition of methanol to the oxonium ion (breaking the π bond and not the σ bond, of course).
4. Loss of a proton to give the acetal.

Oxonium ions

Oxonium ions have three bonds to a positively charged oxygen atom. All three bonds can be σ bonds, as in H_3O^+ or Meerwein's salt, trimethyloxonium fluoroborate, a stable (though reactive) alkylating agent, or one bond can be a π bond as in the acetal intermediate. The term 'oxonium ion' describes either of these structures. They are like alkylated ethers or *O*-alkylated carbonyl compounds.



Just as protonated carbonyl groups are much more electrophilic than unprotonated ones, these oxonium ions are powerful electrophiles. They can react rapidly with a second molecule of alcohol to form the new, stable compounds known as acetals. An oxonium ion was also an intermediate in the formation of hemiacetals in acid solution. Before reading any further, it would be worthwhile to write out the whole mechanism of acetal formation from aldehyde or ketone plus alcohol through the hemiacetal to the acetal, preferably without looking at the fragments of mechanism above or the answer overleaf.

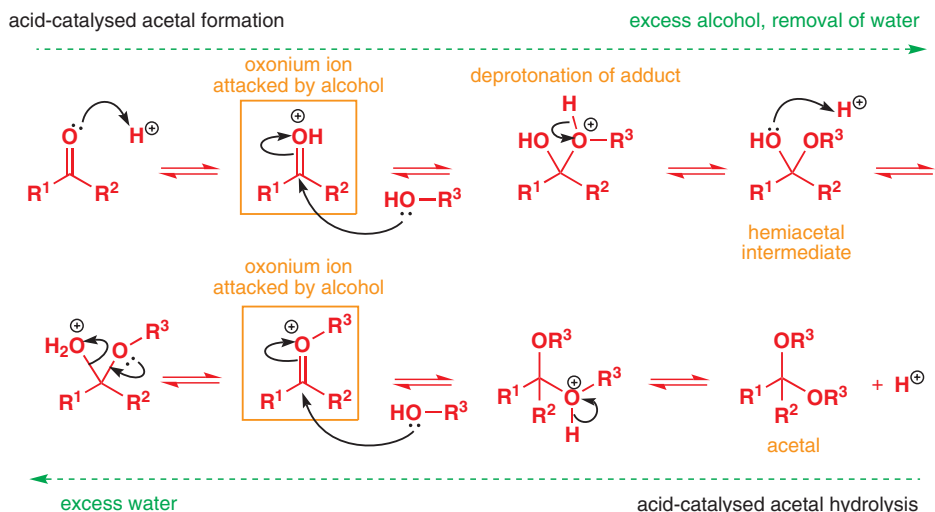
● Formation of acetals and hemiacetals

Hemiacetal formation is catalysed by acid or base, but acetal formation is possible only with an acid catalyst because an OH group must be made into a good leaving group.



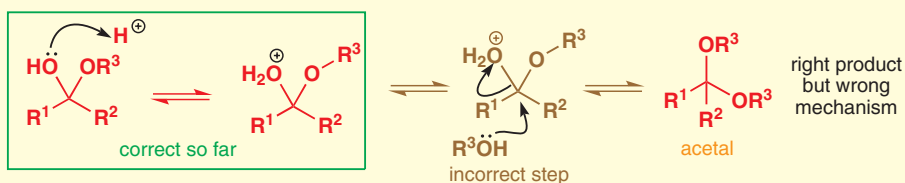
The mechanism is the most complex you have met and it will help you to recall it if you see it in two halves, each very similar to the other. The reaction starts with a protonation on carbonyl oxygen and addition of an alcohol to the $\text{C}=\text{O}$ π bond. When you get to the temporary haven of the hemiacetal, you start again with protonation of that same oxygen then lose the OH group by breaking what was the $\text{C}=\text{O}$ σ bond to form an oxonium ion. Each half goes through an oxonium ion and the alcohol adds to each oxonium ion. The last step in the formation of both the acetal and the hemiacetal is the loss of a proton from the recently added alcohol. From your complete mechanism you should also be able to verify that acetal formation is indeed catalytic in acid.

Interactive mechanism for acetal formation



• Remember the oxonium ion!

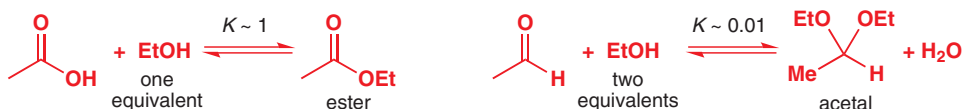
When you wrote out your mechanism for acetal formation, we hope you didn't miss out the oxonium ion! It's easy to do so, but the mechanism most definitely does not go via a direct displacement of water by alcohol.



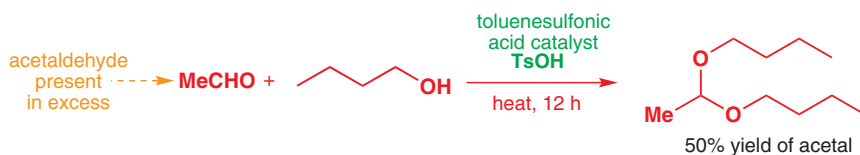
If you wonder how we know this, consult a specialized book on organic reaction mechanisms. After you have read Chapter 15 in this book, you will be able to spot that this substitution step goes via an S_N1 and not an S_N2 mechanism.

Making acetals

Just as with the ester formation and hydrolysis reactions we discussed in Chapter 10, every step in the formation of an acetal is reversible. To make acetals, therefore, we must use an excess of alcohol or remove the water from the reaction mixture as it forms, by distillation for example.

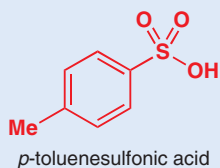


In fact, acetal formation is even more difficult than ester formation: while the equilibrium constant for acid-catalysed formation of ester from carboxylic acid plus alcohol is usually about 1, for acetal formation from an aldehyde and ethanol (shown above), the equilibrium constant is $K = 0.0125$. For ketones, the value is even lower: in fact, it is often very difficult to make the acetals of ketones (sometimes called ketals) unless they are cyclic (we consider cyclic acetals later in the chapter). However, there are several techniques that can be used to prevent the water produced in the reaction from hydrolysing the product.

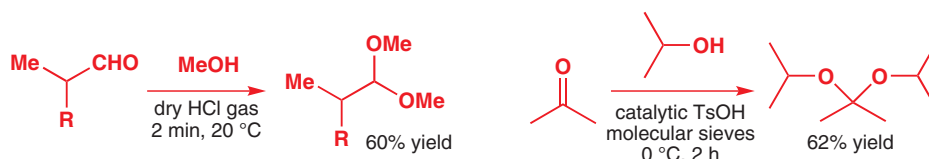


para-Toluenesulfonic acid

para-Toluenesulfonic acid is commonly used to catalyse reactions of this sort. It is a stable solid, yet is as strong an acid as sulfuric acid. It is widely available and cheap because it is produced as a by-product in the synthesis of saccharin (for more details, see Chapter 21).



With the more reactive aldehyde, it was sufficient just to have an excess of one of the reagents (acetaldehyde) to drive the reaction to completion. Dry HCl gas can work too. With a less reactive ketone, molecular sieves (zeolite) were used to remove water from the reaction as it proceeded.



Molecular sieves are minerals that have very small cavities that can absorb only even smaller molecules. The ones used in acetal formation selectively absorb water. They are supplied as tiny cylinders of whitish material.

Acetals hydrolyse only in the presence of acid

Just as acetal formation requires acid catalysis, acetals can be hydrolysed only by using an acid catalyst. With aqueous acid, the hydrolysis of acyclic acetals is very easy. Our examples are the two acetals we made earlier.

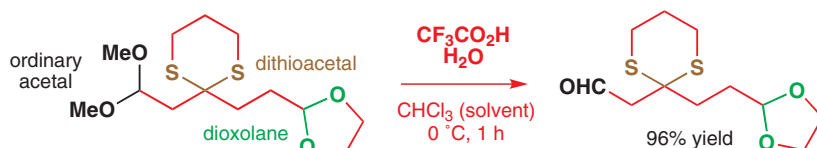
**● Acetal hydrolysis**

Acetals can be hydrolysed in acid but are stable to base.

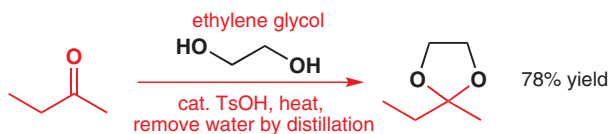
We won't go through the mechanism again—you've already seen it as the reverse of acetal formation, but the fact that acetals are stable to base is really a very important point, which we will use on the next page and capitalize on further in Chapter 23.

Cyclic acetals are more stable than acyclic acetals

Of course you want us to prove it. Well, in this example the starting material has three acetals: an ordinary acetal formed from methanol (in black), a five-membered cyclic acetal, and a dithioacetal. Only the black acetal hydrolyses under these mild conditions.



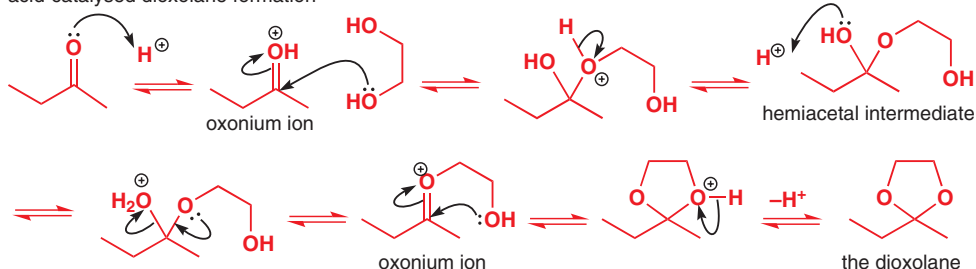
The acetals you have met so far were formed by reaction of two molecules of alcohol with one of carbonyl compound. Cyclic acetals, formed by reaction of a single molecule of a diol, a compound containing two hydroxyl groups, are also important. When the diol is ethylene glycol (as in this example) the five-membered cyclic acetal is known as a **dioxolane**.



■ We hope you didn't make the mistake of missing out the oxonium ion steps!

Before looking at the answer below, try to write a mechanism for this reaction. If you need it, use the mechanism we gave for the formation of acyclic acetals.

acid-catalysed dioxolane formation



Interactive mechanism for cyclic acetal formation

■ Cyclic acetals like this are more resistant to hydrolysis than acyclic ones, and easier to make—they form quite readily even from ketones. One explanation for this is that whenever the second oxonium ion in this mechanism forms, the hydroxyl group is always held close by, ready to snap shut and give back the dioxolane; water gets less of a chance to attack it and hydrolyse the acetal. We will discuss in *entropic* terms why cyclic acetals and hemiacetals are more stable in Chapter 12.

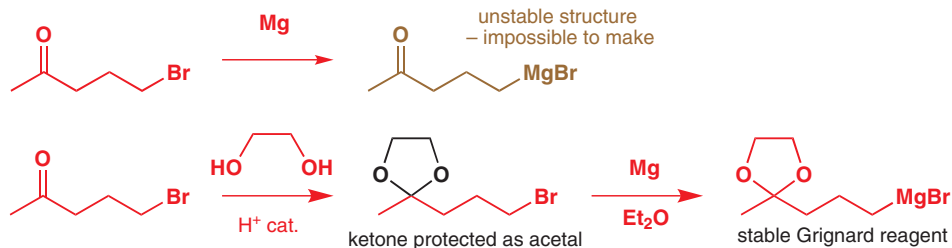
Water is still generated, and needs to be got rid of: in the example above you can see that water was distilled out of the reaction mixture. This is possible with these diols because they have a boiling point above that of water (the boiling point of ethylene glycol is 197 °C). You can't distil water from a reaction mixture containing methanol or ethanol because the alcohols distil too! One very useful piece of equipment for removing water from reaction mixtures containing only reagents that boil at higher temperatures than water is called a Dean Stark head.

Dean Stark head

When a mixture of toluene and water boils, the vapour produced is a constant ratio mixture of toluene vapour and water vapour known as an **azeotrope**. If this mixture is condensed, the liquid toluene and water, being immiscible, separate out into two layers with the water below. By using a Dean Stark apparatus, or Dean Stark head, the toluene layer can be returned to the reaction mixture while the water is removed. Reactions requiring removal of water by distillation are therefore often carried out in refluxing toluene or benzene under a Dean Stark head.

Modifying reactivity using acetals

Why are acetals so important? Well, they're important to both nature and chemists because many carbohydrates are acetals or hemiacetals (see the box below). One important use that chemists have put them to is as *protecting groups*. One synthesis of the steroid class of compounds (about which more later) requires a Grignard reagent with an impossible structure. This compound cannot exist as the Grignard functional group would attack the ketone: it would react with itself. Instead, the protected Grignard reagent is used, made from the same bromoketone, but with an acetal-forming step.

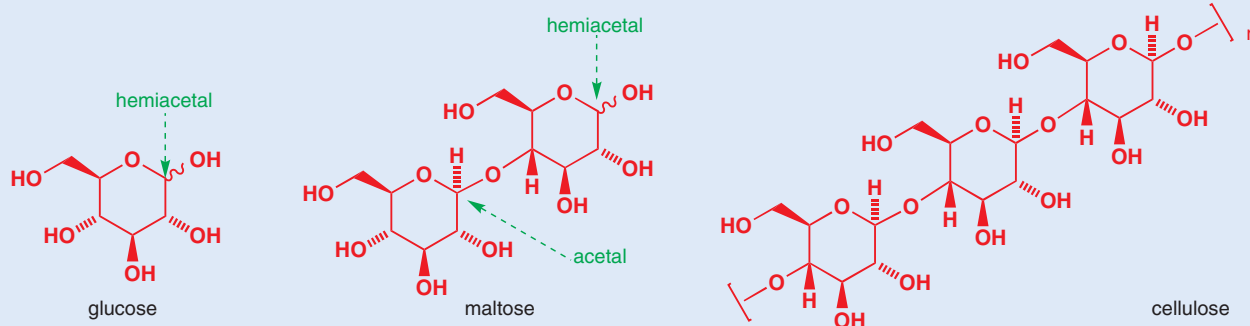


Acetals, as we stressed, are stable to base and to basic nucleophiles such as Grignard reagents, so we no longer have a reactivity problem. Once the Grignard reagent has reacted with an electrophile, the ketone can be recovered by hydrolysing the acetal in dilute acid. The acetal is functioning here as a protecting group because it protects the ketone from attack by the Grignard reagent. Protecting groups are extremely important in organic synthesis and we will return to them in Chapter 23.

Acetals in nature

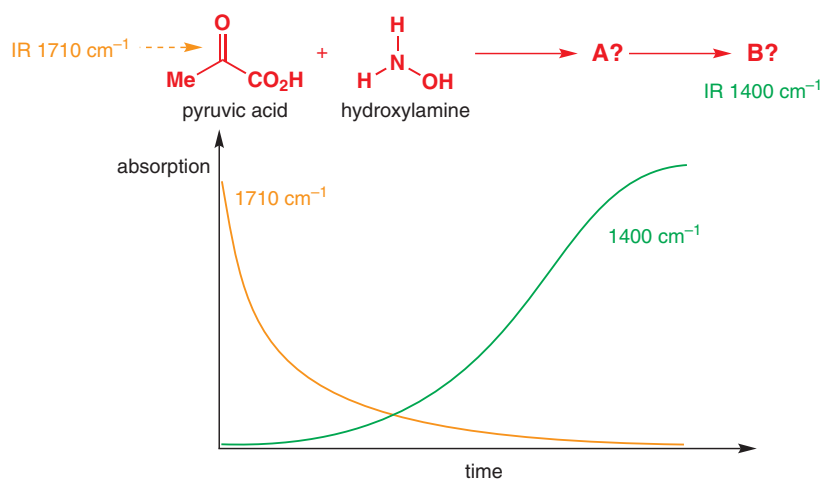
We showed you glucose as on p.137 an example of a stable, cyclic hemiacetal. Glucose can, in fact, react with itself to form an acetal known as maltose. Maltose is a disaccharide (made of two sugar units) produced by the enzymatic

hydrolysis of starch or cellulose, which are themselves polyacetals made up of a string of glucose units.



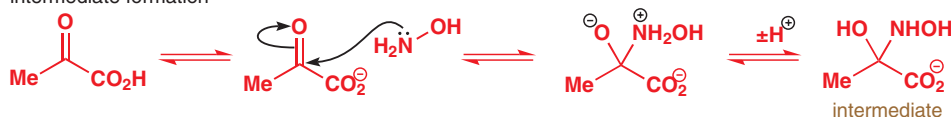
Amines react with carbonyl compounds

The ketone carbonyl group of pyruvic acid (or 2-oxopropanoic acid) has a stretching frequency of a typical ketone, 1710 cm^{-1} . When hydroxylamine is added to a solution of pyruvic acid, this stretching frequency slowly disappears. Later, a new IR absorption appears at 1400 cm^{-1} . What happens?



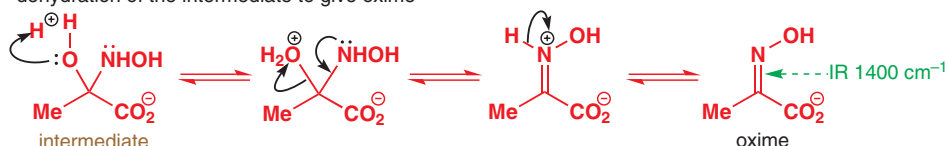
You can probably apply something of what you know from Chapters 6 and 10 about the reactivity of carbonyl compounds towards nucleophiles to work out what is happening in this reaction between a carbonyl compound and an amine. The hydroxylamine first adds to the ketone to form an unstable intermediate similar to a hemiacetal.

intermediate formation

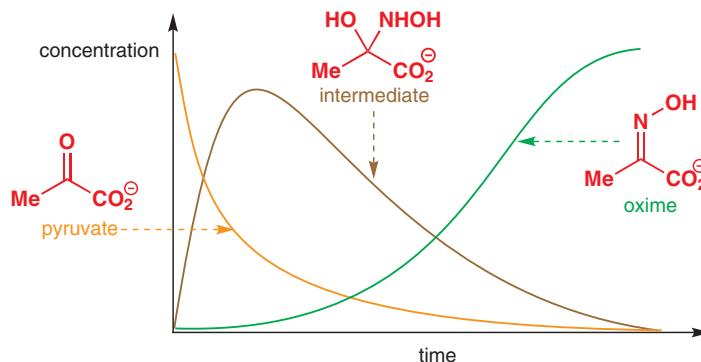


Notice that it is the more nucleophilic nitrogen atom, and not the oxygen atom, of hydroxylamine that adds to the carbonyl group. Like hemiacetals, these intermediates are unstable and can decompose by loss of water. The product is known as an oxime and it is this compound, with its $\text{C}=\text{N}$ double bond, that is responsible for the IR absorption at 1400 cm^{-1} .

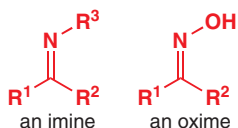
dehydration of the intermediate to give oxime



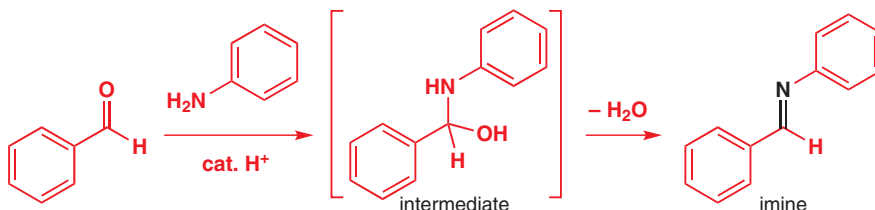
We know that the oxime is formed *via* an intermediate because the 1400 cm^{-1} absorption hardly appears until after the 1710 cm^{-1} absorption has almost completely gone. There must really be another curve to show the formation and the decay of the intermediate. The only difference is that the intermediate has no double bond to give an IR absorbance in this region of the spectrum. We come back to oximes later in the chapter.



Imines are the nitrogen analogues of carbonyl compounds

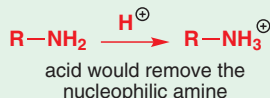


In fact, the oxime formed from a ketone and hydroxylamine is just a special example of an imine. All imines have a C=N double bond and are formed when any primary amine reacts with an aldehyde or a ketone under appropriate conditions, for example aniline and benzaldehyde.

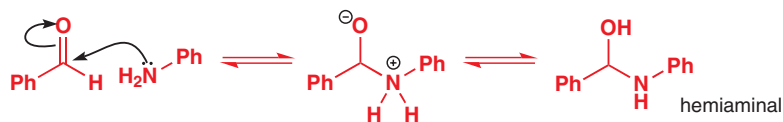


You shouldn't need us to tell you the mechanism of this reaction: even without looking at the mechanism we gave for the formation of the oxime it should come as no surprise to you by now. But as the reaction is very important in chemistry and biology, we'll discuss it in some depth. First, the amine attacks the aldehyde and the intermediate known as a hemiacetal is formed. Amines are good nucleophiles for carbonyl groups, and aldehydes and ketones are electrophilic. There is no need for any catalysis in this step. Indeed, addition of acid would slow the reaction down as the nucleophilic amine would be removed as a salt.

■ Acid would protonate the amine and remove it from the equilibrium and so slow this step down. Acid is not needed for the first step.

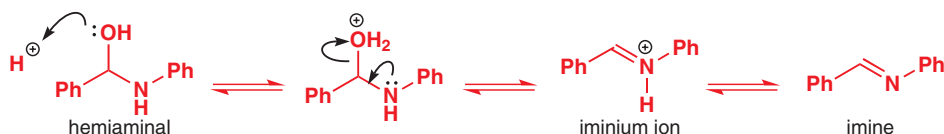


First step in imine formation:
the amine attacks the carbonyl group to form the hemiaminal intermediate:



Dehydration of the hemiaminal gives the imine. Now there is some need for catalysis: acid must be added so that the OH group can become a good leaving group. This step resembles the conversion of hemiacetals to acetals. The difference is that the iminium ion can lose a proton and become a neutral imine.

Second step in imine formation: acid-catalysed dehydration of the hemiaminal intermediate:

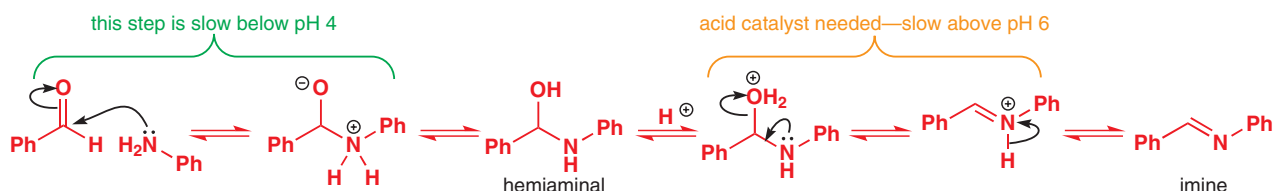


Interactive mechanism for imine formation

● Imine formation requires acid catalysis.

So acid is needed for the second step but hinders the first step. Clearly some compromise is needed. Without an acid catalyst, the reaction is very slow, although in some cases it may still take place. Imine formation is in fact fastest at about pH 4–6: at lower pH, too much amine is protonated and the rate of the first step is slow; above this pH the proton concentration is too low to allow protonation of the OH leaving group in the dehydration step. Imine formation is like a biological reaction: it is fastest near neutrality.

➔ The dependence of the rate of reactions on pH is discussed further in Chapter 12.



Imines are usually unstable and are easily hydrolysed

Like acetals, imines are unstable with respect to their parent carbonyl compound and amine, and must be formed by a method that allows removal of water from the reaction mixture.

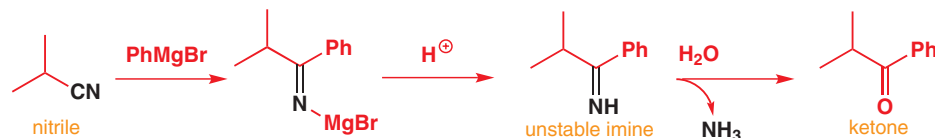


Imines are formed from aldehydes or ketones with most primary amines. In general, they are stable enough to be isolated only if either the C or N of the imine double bond bears an aromatic substituent. Imines formed from ammonia are unstable, but can be detected in solution. $\text{CH}_2=\text{NH}$, for example, decomposes at temperatures above -80°C , but $\text{PhCH}=\text{NH}$ is detectable by UV spectroscopy in a mixture of benzaldehyde and ammonia in methanol.

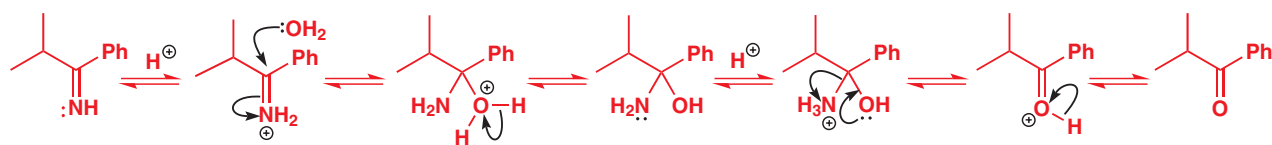


■ Because it is made from an unsymmetrical ketone this imine can exist as a mixture of *E* and *Z* isomers, just like an alkene. When it is formed by this method the ratio obtained is 8:1 *E:Z*. Unlike the geometrical isomers of alkenes, however, those of an imine usually interconvert quite rapidly at room temperature. The geometrical isomers of oximes on the other hand are stable and can even be separated.

Imines are readily hydrolysed to the carbonyl compound and amine by aqueous acid—in fact, except for the particularly stable special cases we discuss below, most can be hydrolysed by water without acid or base catalysis. You have, in fact, already met an imine hydrolysis: at the end of Chapter 10 we talked about the addition of Grignard reagents to nitriles. The product is an imine that hydrolyses in acid solution to ketone plus ammonia.

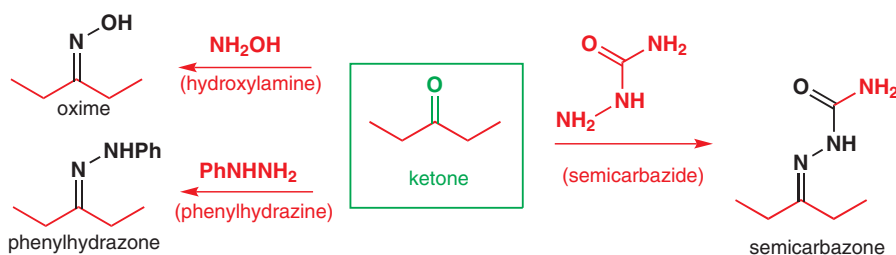


The mechanism of the hydrolysis is the reverse of imine formation, going through the same hemiaminal intermediate and the same iminium and oxonium ions. All these steps are reversible and this should remind you that the relative stability of the starting material and product is as important in imine formation and hydrolysis as it is in acetal formation and hydrolysis.



Some imines are stable

Imines in which the nitrogen atom carries an electronegative group are usually stable: examples include oximes, hydrazones, and semicarbazones.



Interactive mechanism for hydrazone formation

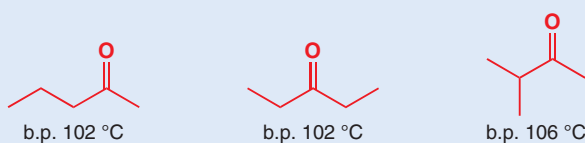
These compounds are more stable than imines because the electronegative substituent can participate in delocalization of the imine double bond. Delocalization decreases the small positive charge on the carbon atom of the imine double bond and raises the energy of the LUMO, making it less susceptible to nucleophilic attack. Oximes, hydrazones, and semicarbazones require acid or base catalysis to be hydrolysed.



Historical note

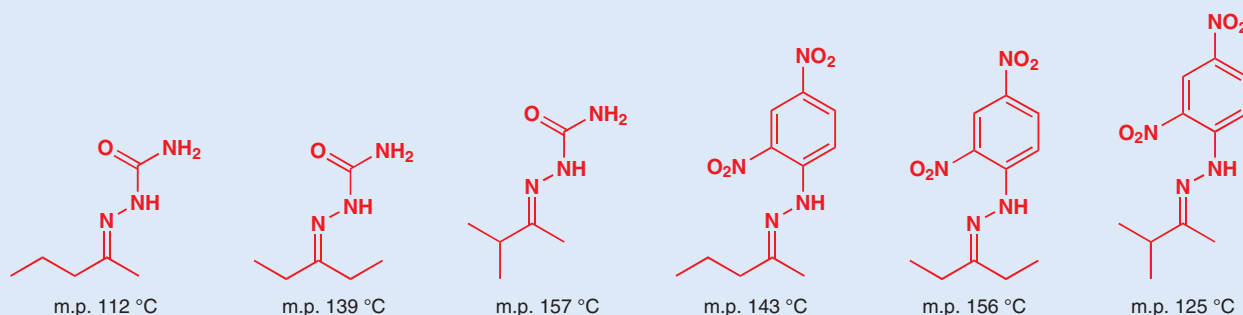
Because the hydrazone and semicarbazone derivatives of carbonyl compounds are often stable, crystalline solids, they used to be used to confirm the supposed identity of aldehydes and ketones. For example, the boiling points of these

three isomeric five-carbon ketones are all similar and before the days of NMR spectroscopy it would have been hard to distinguish between them.



Their semicarbazones and 2,4-dinitrophenylhydrazones, on the other hand, all differ in their melting points. By making these derivatives of the ketones, identification was made much easier. Of course, all of this has been totally super-

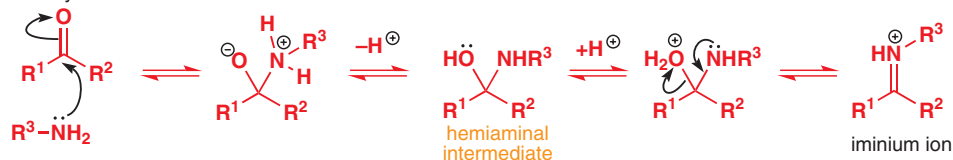
sed by NMR! However, these crystalline derivatives are still useful in the purification of volatile aldehydes and ketones, and in solving structures by X-ray crystallography.



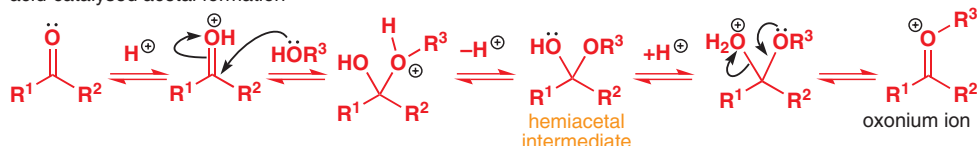
Iminium ions and oxonium ions

Let's return to the mechanism of imine formation, and compare it for a moment with that of acetal formation. The only difference to begin with is that there is no need for acid catalysis for the addition of the amine but there is need for acid catalysis in the addition of the alcohol, a much weaker nucleophile.

acid-catalysed imine formation

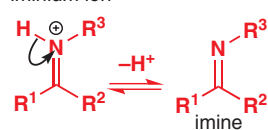


acid-catalysed acetal formation

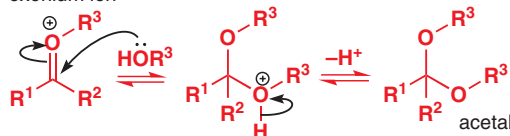


Up to this point, the two mechanisms follow a very similar path, with clear analogy between the hemiaminal and hemiacetal intermediates, and between the iminium and oxonium ions. Here, though, they diverge, because the iminium ion carries a proton, which the oxonium ion doesn't have. The iminium ion therefore acts as an acid, losing a proton to become the imine. The oxonium ion, on the other hand, acts as an electrophile, adding another molecule of alcohol to become the acetal.

iminium ion



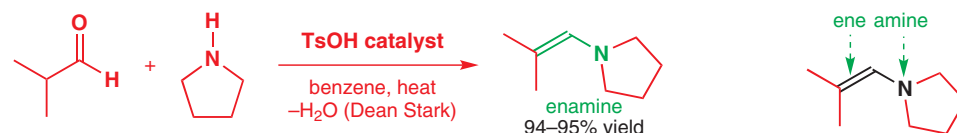
oxonium ion



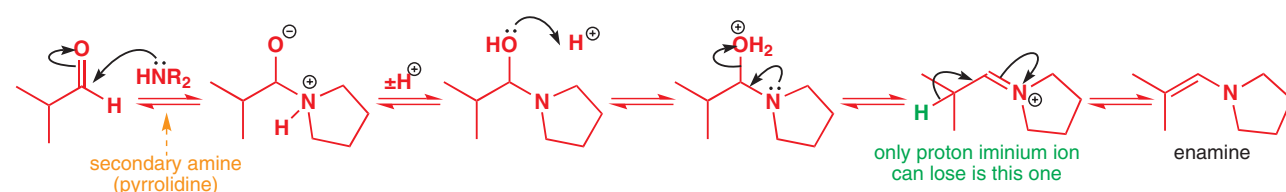
As you might guess, however, iminium ions can be persuaded to act as electrophiles, just like oxonium ions, provided a suitable nucleophile is present. We will spend the next few pages considering reactions in which an iminium ion acts as an electrophile. First, though, we will look at a reaction in which the iminium ion cannot lose an N–H proton because it has none.

Secondary amines react with carbonyl compounds to form enamines

Pyrrolidine, a secondary amine, reacts with isobutyraldehyde, under the sort of conditions you would use to make an imine, to give an enamine. The name enamine combines 'ene' (C=C double bond) and 'amine'.



The mechanism consists of the same steps as those that take place when imines form from primary amines, up to formation of the iminium ion. This iminium ion has no N–H proton to lose, so it loses one of the C–H protons next to the C=N to give the enamine. Enamines, like imines, are unstable to aqueous acid.



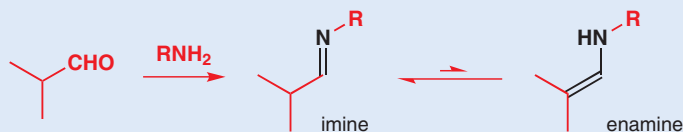
 Interactive mechanism for enamine formation

● Imines and enamines

- Imines are formed from aldehydes or ketones with primary amines.
- Enamines are formed from aldehydes or ketones with secondary amines.

Both require acid catalysis and removal of water.

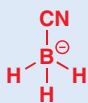
Enamines of primary amines, or even of ammonia, also exist, but only in equilibrium with an imine isomer. The interconversion between imine and enamine is the nitrogen analogue of **enolization**, which is discussed in detail in Chapter 20.



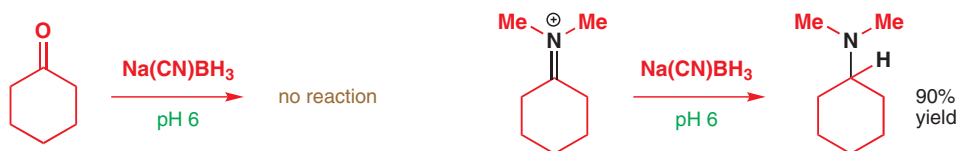
Iminium ions can react as electrophilic intermediates

We made the point above that the difference in reactivity between an iminium ion and an oxonium ion is that an iminium ion can lose H^+ and form an imine or an enamine, while an oxonium ion reacts as an electrophile. Iminium ions can, however, react as electrophiles provided suitable nucleophiles are present. In fact, they are very good electrophiles, and are significantly more reactive than carbonyl compounds. For example, iminium ions are reduced rapidly by the mild reducing agent sodium cyanoborohydride, $Na(CN)BH_3$, while carbonyl compounds are not. An alternative to $Na(CN)BH_3$ is $NaBH(OAc)_3$ (sodium triacetoxyborohydride)—somewhat safer because strong acid can release deadly HCN from $Na(CN)BH_3$.

Sodium cyanoborohydride contains the cyanoborohydride anion, whose structure has tetrahedral boron.

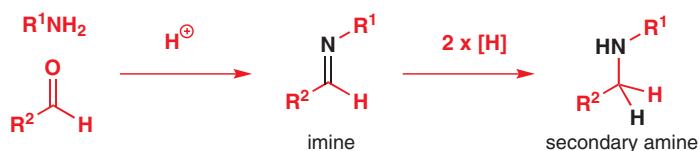


It is a 'toned down' version of sodium borohydride—the electron-withdrawing cyano group decreases the ease with which hydride is transferred.

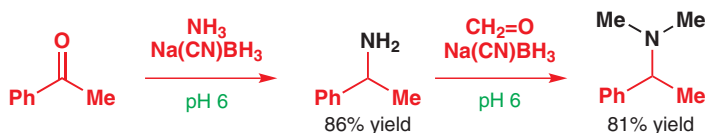


Amines from imines: reductive amination

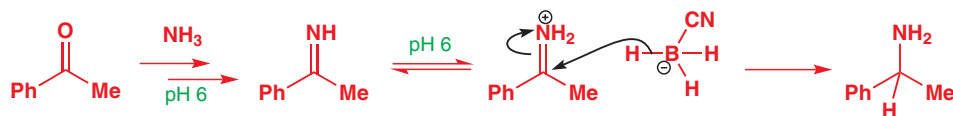
A useful way of making amines is by reduction of imines (or iminium ions). This overall process, from carbonyl compound to amine, is called **reductive amination**. This is, in fact, one of the few successful ways, and the best way, of making secondary amines. This should be your first choice in amine synthesis.



This can be done in two steps, provided the intermediate is stable, but, because the instability of many imines makes them hard to isolate, the most convenient way of doing it is to form and reduce the imine in a single reaction. The selective reduction of iminium ions (but not carbonyl compounds) by sodium cyanoborohydride makes this possible. When $Na(CN)BH_3$ is added to a typical imine-formation reaction it reacts with the iminium ion but not with the starting carbonyl compound nor with the imine. Here is an example of an amine synthesis using reductive amination.

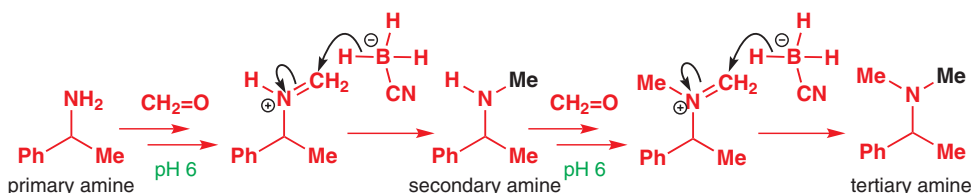


In the first step, the ketone and ammonia are in equilibrium with their imine, which, at pH 6, is partly protonated as an iminium ion. The iminium ion is rapidly reduced by the cyanoborohydride to give the amine. Reactions like this, using ammonia in a reductive amination, are often carried out with ammonium chloride or acetate as convenient sources of ammonia. At pH 6, ammonia will be mostly protonated anyway as the pK_a of NH_4^+ is about 10.



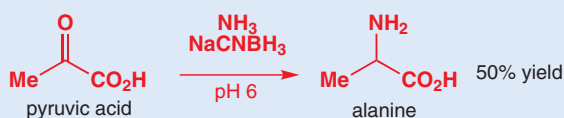
➡ You will again meet the highly electrophilic iminium ions produced by reaction of formaldehyde with amines in Chapter 26, where we introduce you to the Mannich reaction.

In the second step of the synthesis, amine plus formaldehyde gives an imine, present as its protonated iminium form, which gets reduced. Formaldehyde is so reactive that it reacts again with the secondary amine to give an iminium ion; this too is reduced to the amine.



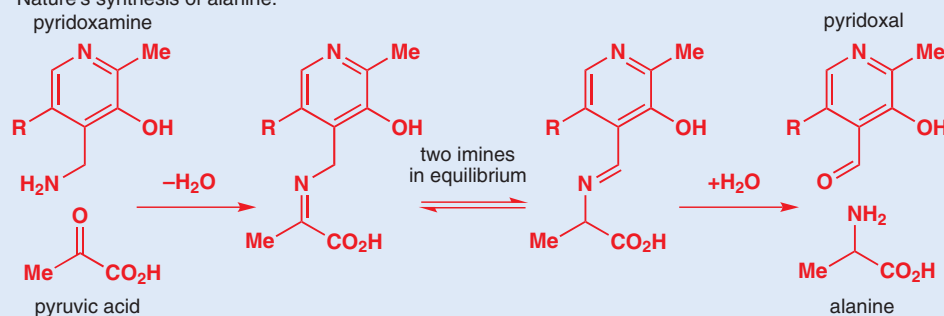
Living things make amino acids using imines

The amino acid alanine can be made in moderate yield in the laboratory by reductive amination of pyruvic acid.



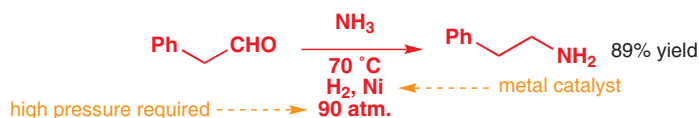
Living things use a very similar reaction to manufacture amino acids from keto acids, but do it much more efficiently. The key step is the formation of an imine between pyruvic acid and the vitamin B₆-derived amine pyridoxamine.

Nature's synthesis of alanine:
pyridoxamine



This imine (biochemists call imines Schiff bases) is in equilibrium with an isomeric imine, which can be hydrolysed to pyridoxal and alanine. These reactions are, of course, all controlled by enzymes and coupled to the degradation of unwanted amino acids (the latter process converts the pyridoxal back to pyridoxamine). Nature was doing reductive aminations a long time before sodium cyanoborohydride was invented! We will come back to this in Chapter 42.

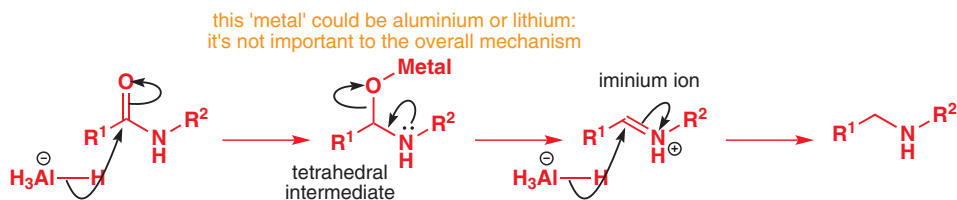
An alternative method for reductive amination uses hydrogenation (hydrogen gas with a metal catalyst) to reduce the imine in the presence of the carbonyl compound. Most of these reductions do not require such high temperatures or pressures.



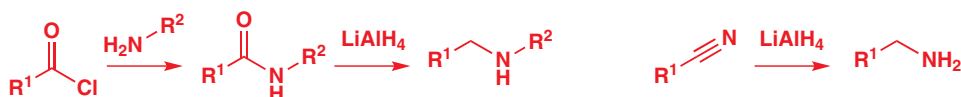
■ Hydrogenation is a good way of reducing a number of different functional groups, but not (usually) carbonyl groups. In Chapter 23 we will look in more detail at reducing agents (and other types of reagent) that demonstrate selectivity for one functional group over another (chemoselectivity).

Lithium aluminium hydride reduces amides to amines

We've talked about reduction of iminium ions formed from carbonyl compounds plus amines. Iminium ions can also be formed by reducing amides with lithium aluminium hydride. A tetrahedral intermediate is formed that collapses to the iminium ion.



The iminium ion is, of course, more electrophilic than the starting amides (amide carbonyl groups are about the least electrophilic of any!), so it gets reduced to the secondary amine. This reaction can be used to make secondary amines from primary amines and acyl chlorides. A similar reduction with lithium aluminium hydride gives a primary amine from a nitrile.



Cyanide will attack iminium ions: the Strecker synthesis of amino acids

Cyanide will react with iminium ions to form α amino nitriles. Although these compounds are relatively unimportant in their own right, a simple hydrolysis step produces α amino acids. This route to amino acids is known as the Strecker synthesis. Of course, it's not usually necessary to make the amino acids that Nature produces for us in living systems: they can be extracted from hydrolysed proteins. This Strecker synthesis is of phenylglycine, an amino acid not found in proteins. Cyanide reacts more rapidly with the iminium ion generated in the first step than it does with the starting benzaldehyde.

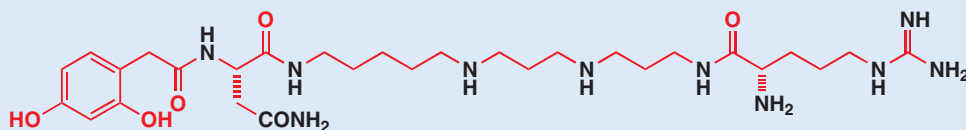
■ Make sure that you can write a mechanism for the hydrolysis of the nitrile to the carboxylic acid! (If you need reminding it is given in Chapter 10.)



The synthesis of a spider toxin: reductive amination

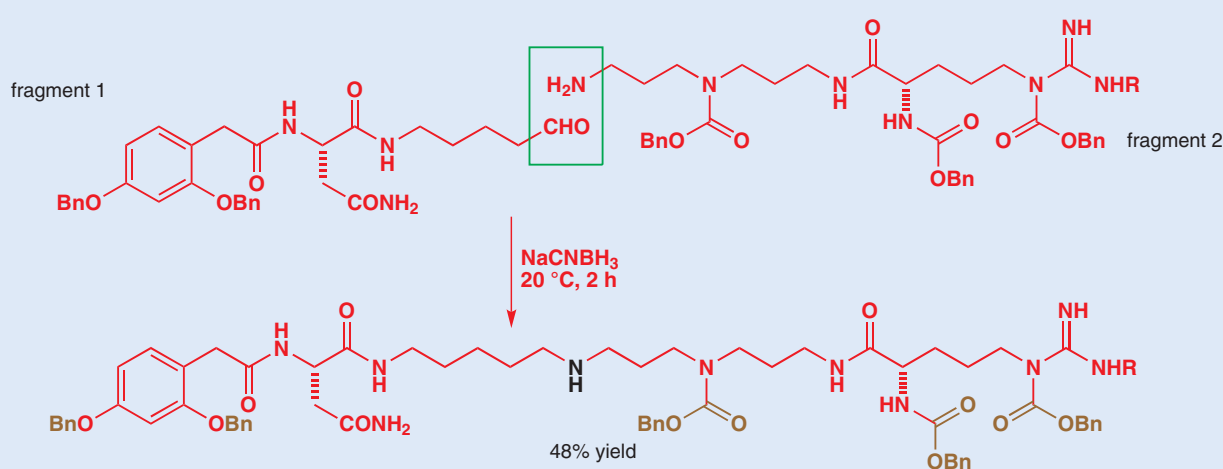
This compound is the toxin used by the orb weaver spider to paralyse its prey. Notice that it has a guanidine at its right-hand end. These are stable imines,

and their powerful basicity was discussed in Chapter 8.



Since the spider produces only minute quantities of the compound, chemists at the University of Bath set about synthesizing it in the laboratory so that they could study its biological properties. The toxin contains several amide and

amine functional groups, and the chemists decided that the best way to make it was to link two molecules together at one of the secondary amine groups using a reductive amination.



The compound made by this reaction has almost, but not exactly, the spider toxin structure. The extra groups in brown are protecting groups that prevent

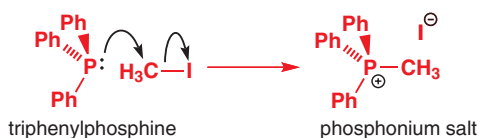
unwanted side-reactions at the other amine and phenol functional groups. We will discuss protecting groups in detail in Chapter 23.

Substitution of C=O for C=C: a brief look at the Wittig reaction

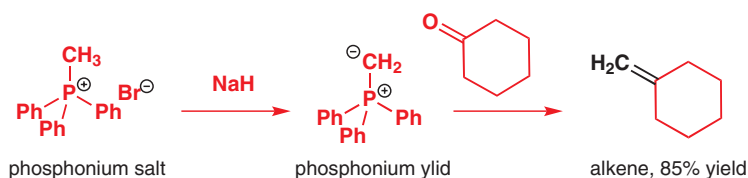
Before we leave substitution reactions of carbonyl groups, there is one more reaction that we must introduce. It is an important one and we will come back to it again later in this book, particularly in Chapter 27. It also has a rather different mechanism from most you have met in recent chapters, but we talk about it here because the overall consequence of the **Wittig reaction** is the substitution of a C=C bond for a C=O bond.

We don't normally tell you the name of a reaction before even mentioning how to do it, but here we make an exception because the reagents are rather unusual and need explaining in detail. The Wittig reaction is a reaction between a carbonyl compound (aldehyde or ketone only) and a species known as a **phosphonium ylid**. An ylid (or ylide) is a species with positive and negative charges on adjacent atoms, and phosphonium ylids are made from **phosphonium** salts by deprotonating them with a strong base.

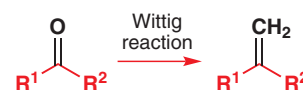
You have already met phosphonium salts in Chapter 5, where you saw the reaction of a phosphine (triphenylphosphine) with an alkyl halide (methyl iodide) to give the tetrahedral phosphonium salt.



So here is a typical Wittig reaction: it starts with a phosphonium salt, which is treated with a strong base such as BuLi or sodium hydride, and then with a carbonyl compound; the alkene forms in 85% yield.



What about the mechanism? We warned you that the mechanism is rather different from all the others you have met in this chapter, but nonetheless it begins with attack on the

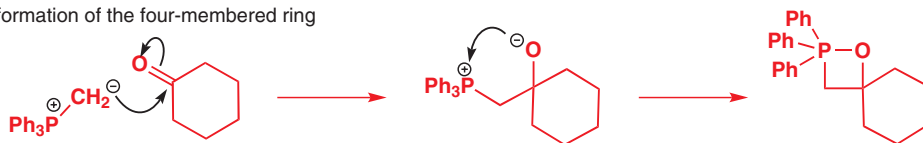


The Wittig reaction is named after its discoverer, the Nobel Prize winner Georg Wittig (1897–1987; Nobel Prize 1979).

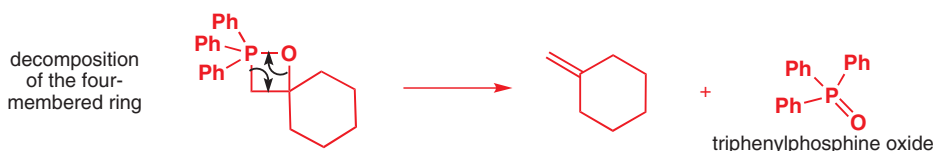
■ The positively charged P atom stabilizes the negative charge on carbon, making phosphonium salts another class of 'carbon acids' (to add to those you met in Chapter 8) that can be deprotonated by strong base. The hydride ion H[−] is the conjugate base of H₂ which has a pK_a of about 35.

carbonyl group by a nucleophile; the nucleophile is the carbanion part of the phosphonium ylid. This reaction generates a negatively charged oxygen that attacks the positively charged phosphorus and gives a four-membered ring called an oxaphosphetane.

formation of the four-membered ring



Now, this four-membered ring (like many others) is unstable, and it can collapse in a way that forms two double bonds. Here are the curly arrows: the mechanism is cyclic and gives the alkene, which is the product of the reaction along with a phosphine oxide.



Interactive mechanism for the Wittig reaction

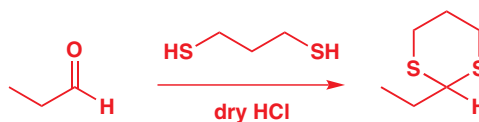
➡ We will look at the Wittig reaction again in more detail in Chapter 27.

The chemistry of some elements is dominated by one particular property, and a theme running right through the chemistry of phosphorus is its exceptional affinity for oxygen. The P=O bond, with its bond energy of 575 kJ mol^{-1} , is one of the strongest double bonds in chemistry, and the Wittig reaction is irreversible and is driven forward by the formation of this P=O bond. No need here for the careful control of an equilibrium necessary when making acetals or imines.

Summary

In this chapter, as in Chapter 10, you have met a wide variety of reactions, but we hope you have again been able to see that they are all related mechanistically. Of course, we have not been exhaustive: it would be impossible to cover every possible reaction of a carbonyl group, but having read Chapters 6, 9, and 10 you should feel confident in writing a *reasonable mechanism* for any reaction involving nucleophilic attack on a carbonyl group. You could try thinking about this, for example.

■ Hint. Consider sulfur's location in the periodic table.



In the next chapter we examine in a little more detail the phrase 'a reasonable mechanism': how do we know what mechanisms are reasonable, and what can we do to understand them? We shall look in more detail at some of the topics raised in this chapter, such as equilibria and rates of reactions. Carbonyl groups next star in Chapter 20 where they reveal a thus far hidden *nucleophilic* side to their character.

Further reading

Section 3, 'Nucleophilic substitution to the carbonyl group with complete removal of carbonyl oxygen', in S. Warren, *Chemistry of the Carbonyl Group*, Wiley, Chichester, 1974.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>